5.

	Type of conic	Centre	Vertices	Foci	Directrices
i.	Ellipse	$(0,0)$	$(\pm5,0)$	$(\pm4,0)$	$x=\pm\frac{25}{4}$
ii.	Ellipse	$(0,0)$	$\left(0,\pm\sqrt{10}\right)$	$\left(0,\pm\sqrt{7}\right)$	$y=\pm\frac{10}{\sqrt{7}}$
iii.	Hyperbola	$(0,0)$	$(\pm5,0)$	$(\pm13,0)$	$x=\pm\frac{25}{13}$
iv.	Hyperbola	$(0,0)$	$(0,\pm4)$	$(0,\pm5)$	$y=\pm\frac{16}{5}$

8.

	Type of Conic	Centre	Vertices	Foci	Directrices
i.	Ellipse	$(3,4)$	$(3,21),(3,-13)$	$(3,12),(3,-4)$	$y=\frac{321}{8}$, $y=\frac{-257}{8}$
ii.	Ellipse	$(-1,2)$	$(-11,2),(9,2)$	$(-7,2),(5,2)$	$x=\frac{47}{3}$, $x=\frac{-53}{3}$
iii.	Hyperbola	$(-3,4)$	$(-18,4),(12,4)$	$(-20,4),(14,4)$	$x=\frac{174}{17}$, $x=\frac{-276}{17}$
iv.	Hyperbola	$(-1,2)$	$(-1,7),(-1,-3)$	$(-1,2+\sqrt{41})$ $(-1,2-\sqrt{41})$	$y=\frac{25}{\sqrt{41}}+2$, $y=\frac{-25}{\sqrt{41}}+2$
v.	Ellipse	$(4,-2)$	$\left(4,-2+3\sqrt{2}\right)$, $(4,-2-3\sqrt{2})$	$(4,-2+\sqrt{6})$, $(4,-2-\sqrt{6})$	$y=-2+3\sqrt{6}$, $y=-2-3\sqrt{6}$
vi.	Hyperbola	$(2,-3)$	$(3,-3),(1,-3)$	$(2+\sqrt{10}-3)$, $(2-\sqrt{10},-3)$	$x=\frac{1}{\sqrt{10}}+2$, $x=\frac{-1}{\sqrt{10}}+2$

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